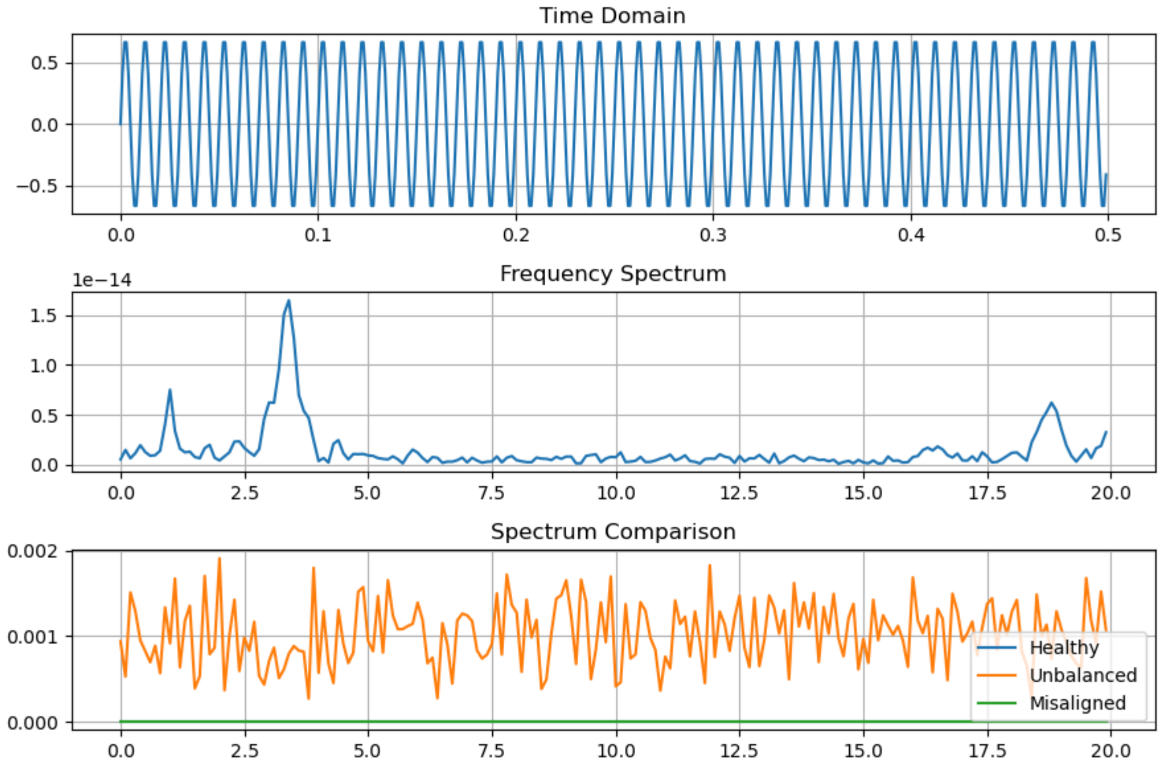
In [10]:
```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
from scipy.fft import fft,fftfreq

import os
path=os.path.join(os.path.expanduser("~"),"Desktop","338","vibrations (1).csv")

df=pd.read_csv(path)
print(df.head())

x=df.select_dtypes(include='number').iloc[:,0].values
fs=1000
t=np.arange(len(x))/fs
N=len(x)
f=fftfreq(N,1/fs)[:N//2]
F=lambda s: 2/N*np.abs(fft(s)[:N//2])
u=x+0.5*np.sin(2*np.pi*25*t) #Unbalanced
m=x=0.7*np.sin(2*np.pi*100*t) #Misaligned

plt.figure(figsize=(9,6))
plt.subplot(3,1,1);plt.plot(t[:500],x[:500]);plt.title("Time Domain");plt.grid()
plt.subplot(3,1,2);plt.plot(f[:200],F(x)[:200]);plt.title("Frequency Spectrum")
plt.grid()
plt.subplot(3,1,3)
plt.plot(f[:200],F(x)[:200],label="Healthy")
plt.plot(f[:200],F(u)[:200],label="Unbalanced")
plt.plot(f[:200],F(m)[:200],label="Misaligned")
plt.legend();plt.title("Spectrum Comparison");plt.grid()
plt.tight_layout();plt.show()

rms=np.sqrt(np.mean(x**2))
cond="HEALTHY" if rms<0.5 else"UNBALANCED" if rms<1 else"MISALIGNED/FAULTY"
print("\nVIBRATION ANALYSIS RESULT")
print("RMS Value:",round(rms,4))
print("Engine Condition:",cond)
```

```
         mm
0 -1.031900
1  0.651533
2 -0.581850
3 -0.538830
4  0.971045
```




```
VIBRATION ANALYSIS RESULT
RMS Value: 0.495
Engine Condition: HEALTHY
```

In [ ]: